MIL-P-47271A(MI)
18 February 1983
SUPERSEDING
MIL-P-47271(MI)
9 August 1974

MILITARY SPECIFICATION

PAPER, INSULATING, ELECTRICAL

This specification is approved for use by the US Army Missile Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers natural colored electrical insulating papers which are not sized, loaded, or dyed.

1.2 Classification.

1.2.1 Grades. The electrical insulating papers shall be of the following grades, as specified:

Grade A - High density, relatively nonporous and relatively nonabsorbent

Grade B - Medium density, relatively nonporous and relatively nonabsorbent

Grade C - Low density, porous and absorbent

1.2.2 Classes. Grade A insulating paper shall be of the following classes, as specified:

Class 1 - nonwaxed

Class 2 - waxed

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Missile Command, ATTN: DRSMI-RSDS, Redstone Arsenal, AL 35898, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5970

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* 1.3 Military part number. The military part number shall consist of the letter B, the basic number of this specification, and a dash number compiled from the classification and thickness required:

Example:	B47271-	A	*2	C
	Basic Number of MIL SPEC	Grade (1.2.1)	Class (1.2.2)	Thickness (Tables II & III) (see 6.3 for Code)

*NOTE: When ordering Grades B or C, a dash (-) shall be used in lieu of the class number, (e.g. B47271-B-S).

2. APPLICABLE DOCUMENTS

* 2.1 Government documents.

* 2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

UU-P-31	Paper, General Specifications and Methods of Testing.
UU-P-271	Paper, Wrapping, Waterproofed, Kraft.

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STANDARDS

Federal

FED-STD-406	Plastic, Methods of Testing

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-1188	Commercial Packaging of Supplies and Equipment

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*(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

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* 2.2 Other publications. The following document(s) forms a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

* American Society for Testing and Materials

ASTM D149	Dielectric Breakdown Voltage and Dielectric Strength for Electrical Insulating Materials at Commercial Power Frequencies, Methods of Test for
ASTM D202	Sampling and Testing Untreated Paper Used for Electrical Insulation
ASTM D646	Thickness of Paper and Paper Products, Methods of Test for
ASTM D726	Resistance of Paper to Passage of Air, Methods of Test for
ASTM D774	Bursting Strength of Paper, Method of Test for
ASTM D1305	Electrical Insulating Paper - Interlayer Type

* (Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

*

* (Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

* 2.3 Order of precedence. In the event a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

* 3.1 First article sample. Unless otherwise specified (see 6.2), a first article sample shall meet all the requirements of this specification.

3.2 Material. The material shall be an unbleached sulfate paper made from spruce or equivalent coniferous woodpulp.

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3.3 Color. Unless otherwise specified, the color of the paper shall be natural tan. Pigments or dyes shall not be used for coloring.

3.4 Chemical requirements. The insulating papers shall meet the requirements as specified in Table I.

Table I. Chemical Requirements

Property	Value			
	Grades A and B		Grade C	
	Min.	Max.	Min.	Max.
Ash (percent)	---	0.5	---	1.00
Water soluble material (percent)	---	---	---	1.25
Alcohol soluble material (percent)	---	1.5	---	1.00
Hydrogen ion concentration (pH)	6.5	7.5	6.5	---
Moisture, as received (percent)	5.00	7.00	4.00	7.00
Aqueous extract conductivity (micromhos)	---	15	---	---
Fiber content (percent)	100*	---	100*	---
Chlorides (as Cl) (percent)	---	0.020	---	0.020
*Unbleached sulfate				

3.5 Physical and electrical requirements.

3.5.1 Grades A and B. Grades A and B insulating papers shall meet the physical and electrical requirements as specified in Table II.

3.5.2 Grade C. Grade C paper shall meet the physical and electrical requirements as specified in Tables III and IV.

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Table II - Physical and electrical requirements for Grades A and B

Nominal Thickness (mil.)	Thickness tolerance (mil, plus or minus)		Apparent density (min., grams per cubic centimeter)		Tensil strength (min., lbs. per inch of width)				Conducting paths (max., number per square foot)		Dielectric strength (min., volts per mil)	
					Machine direction		Cross direction					
	Grade A	Grade B	Grade A	Grade B	Grade A	Grade B	Grade A	Grade B	Grade A	Grade B	Grade A	Grade B
0.20	0.02	----	1.00	----	1.5	----	1.0	----	9	----	1250	----
0.25	0.02	----	1.00	----	2.5	----	1.0	----	7	----	1200	----
0.3	0.02	----	1.00	----	3.5	----	1.5	----	7	----	1000	----
0.35	0.03	----	1.00	----	4.0	----	1.8	----	7	----	950	----
0.4	0.03	----	1.00	----	4.5	----	2.0	----	6	----	850	----
0.45	0.03	----	1.00	----	5.0	----	2.5	----	5	----	750	----
0.5	0.03	----	1.00	----	6.0	----	3.0	----	4	----	700	----
0.55	0.03	----	1.00	----	7.0	----	3.5	----	4	----	650	----
0.6	0.03	----	1.00	----	8.0	----	4.0	----	3	----	600	----
0.75	0.04	----	1.00	----	9.0	----	5.0	----	2	----	500	----
1.0	0.05	0.05	1.00	0.83	14.0	13.0	8.0	7.0	2	2	400	350
1.5	0.07	0.07	1.00	0.83	19.0	17.0	10.0	9.0	2	2	400	300
1.7	0.08	0.08	1.00	0.83	21.0	19.0	11.0	10.0	1	1	400	300
2.0	0.10	0.10	1.00	0.83	24.5	23.5	13.0	12.0	1	1	400	300
2.5	0.12	0.12	1.00	0.83	29.0	25.0	15.0	13.0	1	1	375	300
3.0	0.15	0.15	1.00	0.83	34.0	28.0	18.0	14.0	1	1	350	200
4.0	0.20	0.20	1.00	0.83	40.0	36.0	22.0	16.0	1	1	300	175
5.0	0.25	0.25	1.00	0.83	55.0	50.0	26.0	24.0	1	1	250	175
6.0	0.30	0.30	1.00	0.83	65.0	60.0	31.0	28.0	1	1	225	150
10.0	1.0	1.0	1.00	0.83	100.0	90.0	50.0	45.0	1	1	200	150
15.0	1.5	1.5	1.00	0.83	150.0	135.0	65.0	60.0	1	1	175	150
20.0	2.0	2.0	1.00	0.83	200.0	160.0	80.0	70.0	1	1	150	125
30.0	3.0	3.0	1.00	0.83	250.0	200.0	100.0	85.0	1	1	125	100

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* Table III - Physical requirements for Grade C paper

Property	Nominal thickness (inches) (See Table VI)			
	0.004	0.006	0.007	0.010
Bursting strength, dry, individual roll average (minimum, pounds per square inch)	17	25	30	43
Tensile strength, dry, machine direction individual roll average (minimum, pounds per inch of width)	14	21	25	36
Tensile strength, dry, cross machine direction, individual roll average (minimum, pounds per inch of width	7	10	12	17
Tensile strength, wet, machine direction individual roll average (minimum, pound per inch of width)	0.30	0.45	0.55	0.75
Air resistance (range, seconds per 100 cubic centimeters per square inch)	1.0-2.5	2.0-4.0	2.0-4.0	3.0-6.0
Impregnation Side A (maximum, seconds) Side B (maximum, seconds)	 10 17	 15 25	 18 30	 25 43

* Table IV - Electrical requirements for Grade C paper

Property	Value	
	Min.	Max.
Dielectric strength (average volts per mil)	125	---
Water extract conductivity (micromhos per centimeter	---	40

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3.6 Waxed paper (Grade A, Class 2).

* 3.6.1 Type of wax. The impregnating wax shall be a white mineral impregnating wax.

3.6.2 Insulation resistance. The insulation resistance of waxed insulating paper shall average at least 80 megohms and no single value shall be less than 50 megohms.

* 3.7 Dimensional tolerances.

3.7.1 Grade A and B papers. The width and length tolerances for Grade A and B papers shall be as specified in Table V.

* Table V - Grades A and B papers width and length tolerances

Property (Inch)	Tolerance, plus or minus (Inch)
Width:	
Up to 2	0.0156
2 to 10	0.0313
10 and over	0.0625
Length:	
Sheets	0.0625
Rolls (outside diameter)	0.5

3.7.2 Grade C paper. The dimensional tolerance for Grade C paper shall be as specified in Table VI.

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* Table VI - Grade C paper-Dimensional tolerance

Property	Nominal thickness (inch)			
	0.004	0.006	0.007	0.010
Thickness tolerance (inch) plus or minus	0.0005	0.0005	0.0005	0.001
Width tolerance (inch) plus minus	0.125 0	0.125 0	0.125 0	0.125 0
Core tolerance, inside diameter (inch) plus minus	0 0.0625	0 0.0625	0 0.0625	0 0.0625

* 3.8 Basic weight (Grade C paper). The basic weight for Grade C paper shall be as specified in Table VII.

* Table VII - Grade C-Basic weight

Property	Nominal thickness (inch)			
	0.004	0.006	0.007	0.010
Basic weight (25 by 40-500) (pounds)	33.0	55.0	57.0	79.0
Basic weight tolerances (pounds, plus or minus)	2.0	3.0	3.0	4.0

3.9 Workmanship. The insulating paper shall be uniform in thickness, density, formation and texture. It shall be free from welts, wrinkles, creases, holes, slime spots, blisters, tears, cuts, resin, dirt, carbon, rubber, metallic particles, corrosive substance and other foreign materials.

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4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 First article sample. Unless otherwise specified (see 6.2), the first article sample which has been manufactured under the same conditions as those proposed for subsequent units of production shall be submitted for Government approval. The first article sample will be subjected to all examinations and tests specified herein. Unless otherwise specified, the Government will perform the examinations and tests at the contractor's plant. Prior to submitting another first article sample, the contractor shall furnish the Government full particulars concerning previous rejection and action taken to correct the defects. Subsequent units or lots will not be considered for acceptance until approval of the first article sample has been obtained.

4.3 Inspection provisions.

4.3.1 Lot size.

4.3.1.1 Rolls. For the purpose of sampling, a lot shall consist of all electrical insulating paper of one grade, class and thickness offered for inspection at one time.

* 4.3.1.2 Sheets. For the purpose of sampling, a lot shall consist of all the electrical insulating paper of one grade, class and thickness offered for delivery at one time. A unit of sheets shall consist of not more than 100 pounds of electrical insulation contained in one lot.

4.3.2 Sampling.

4.3.2.1 Sampling of rolls for visual and dimensional inspection. The Government inspector shall select from different rolls, chosen at random from each lot, the number of samples shown in Table VIII in accordance with MIL-STD-105. The outer layer of each roll to be sampled shall be discarded and the next 2 linear yards shall be taken as the sample. The remainder of each roll from which a sample is taken shall be returned to the lot.

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4.3.2.2 Sampling of sheets for visual and dimensional inspection. The Government inspector shall select random samples from each lot as shown in Table VIII in accordance with MIL-STD-105. A sample shall consist of one sheet if 2,090 square inches or more in area. Otherwise, a sample shall consist of two sheets. Each sample shall be taken from a different unit of the lot.

4.3.2.3 Sampling of rolls and sheets for tests. For the sample of rolls specified in 4.3.2.1, or sheets specified in 4.3.2.2, the Government inspector shall further select a sample of five rolls or five units of sheets for the tests specified in 4.4.

Table VIII - Sampling for visual and dimensional inspection AQL (approx.) = 1.5 percent defective

Number of rolls in inspection lot	Number of rolls or units in sample	Acceptance number (defectives)	Rejection number (defectives)
40 or under	15	0	1
41 to 110	25	1	2
111 to 300	35	1	2
301 to 500	50	2	3
501 and over	75	3	4

4.3.3 Examination. Each sample selected in accordance with 4.3.2.1 and 4.3.2.2 shall be inspected by the Government inspector to determine compliance with this specification as to texture, welts, wrinkles, material, color, other foreign materials and defects and measurement of dimensions. Any roll or unit in the sample having one or more visual or dimensional defects shall be rejected and if the number of defective rolls or units exceeds the acceptance number shown in Table VIII the lot represented by the sample shall be rejected.

* 4.4 Test methods and procedures.

* 4.4.1 Tests. Tests shall be performed for the following characteristics in accordance with their corresponding test methods listed in Table IX.

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* Table IX - Test procedures

Characteristics	Tests		
	UU-P-31	ASTM	FED-STD-406
Tensile strength	171		
Apparent density		D202	
Thickness	173		
Conducting paths		D149	
Dielectric strength			4031
Ash	202		
Alcohol, soluble material		D202	
Hydrogen ion	200		
Aqueous extract		D202	
Water solubility		D202	
Fiber composition	300		
Moisture content	240		
Chlorides		D1305 Appendix	
Insulation resistance			4041[1]
* Bursting strength		D774	
* Air resistance		D726	
* Impregnation time		D202	
Weight		D646	

[1]Voltage shall be 100 to 110 volts, d.c; readings shall be made after a steady deflection has been obtained, but not exceeding one minute.

* 5. PACKAGING

* 5.1 Preservation, Packing, Unitization, and Marking - Commerical Packaging (see 6.2).

* 5.1.1 Commercial. Preservation, packing, unitization, and marking shall be in accordance with MIL-STD-1188.

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6. NOTES

6.1 Intended use. Unwaxed papers are intended primarily for use in wound apparatus such as transformers and coils or in other apparatus where they serve as electrically insulating spacers. Waxed insulated paper is intended primarily for use as interleaving paper in wound apparatus.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

* b. Military part number (see 1.3).

c. Grade required.

d. Class (when Grade A is specified).

e. Dimensions including thickness (Specify length and width required).

* f. Whether a first article sample is required. (see 3.1)

g. Selection of applicable levels of preservation, packaging, and packing (see 5.1).

* 6.3 Available thickness. The insulating papers covered by this specification are available in the following thicknesses denoted by the symbol X.

P/N Code	Available thickness, (mil)	Grade A, Class 1	Grade A, Class 2	Grade B	Grade C
A	0.20	X	X	-	-
B	0.25	X	X	-	-
C	0.30	X	X	-	-
D	0.35	X	X	-	-
E	0.40	X	X	-	-
F	0.45	X	X	-	-
G	0.50	X	X	-	-
H	0.55	X	X	-	-
I	0.60	X	X	-	-
J	0.75	X	X	-	-
K	1.0	X	X	X	-
L	1.5	X	-	X	-
M	1.7	X	-	X	-
N	2.0	X	-	X	-
P	3.0	X	-	X	-
Q	4.0	X	-	X	X
R	5.0	X	-	X	-
S	6.0	X	-	X	X
T	7.0	X	-	-	X
U	10.0	X	-	X	X
V	15.0	X	-	X	-
W	20.0	X	-	X	-
X	30.0	X	-	X	-

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* 6.4 Changes to previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modification, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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